# Inserting Missing Numbers — Achievers / HOTS

A harder follow-on to `21-inserting-missing-numbers/`: column-direction patterns, magic-square-style grids (row AND column constraints), two-step rules, number pyramids, and a 2×2 product grid. 15 questions, 4-option MCQ, untimed.

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**Q1.** This time look DOWN each column, not across each row. Every column adds up to the same total. Find the missing number.

| | | |
|---|---|---|
| 9 | 4 | 7 |
| 6 | 10 | **?** |
| 5 | 6 | 10 |

A) 20 B) 3 C) 6 D) 5

**Q2.** This time look DOWN each column, not across each row. Every column adds up to the same total. Find the missing number.

| | | |
|---|---|---|
| 8 | **?** | 6 |
| 9 | 5 | 11 |
| 7 | 9 | 7 |

A) 10 B) 7 C) 9 D) 0

**Q3.** Each row follows a TWO-STEP rule — figure it out from the first two rows, then find the missing number in the third row.

| | | |
|---|---|---|
| 3 | 4 | 14 |
| 5 | 2 | 14 |
| 6 | 3 | **?** |

A) 23 B) 13 C) 18 D) 14

**Q4.** Each row follows a TWO-STEP rule — figure it out from the first two rows, then find the missing number in the third row.

| | | |
|---|---|---|
| 3 | 4 | 10 |

| | | |
|---|---|---|
| 5 | 2 | 12 |
| 6 | **?** | 15 |

A) 1 B) 15 C) 13 D) 3

**Q5.** Each row follows a TWO-STEP rule — figure it out from the first two rows, then find the missing number in the third row.

| | | |
|---|---|---|
| 9 | 4 | 15 |
| 11 | 3 | 24 |
| **?** | 6 | 21 |

A) 13 B) 16 C) 11 D) 8

**Q6.** Every row AND every column adds up to the same total. Find the missing number.

| | | |
|---|---|---|
| 2 | 7 | 6 |
| 9 | **?** | 1 |
| 4 | 3 | 8 |

A) 1 B) 8 C) 15 D) 5

**Q7.** In this number pyramid, each number is the SUM of the two numbers directly below it. Find the missing number at the top.

```
          ?
     16     14
  7      9      5
```

A) 28 B) 25 C) 30 D) 33

**Q8.** In each row, Top = Left + (2 × Middle) + Right. Work out the rule from the first two rows, then find the missing Middle number.

| Left | Middle | Right | Top |
|---|---|---|---|
| 3 | 4 | 5 | 16 |
| 6 | 2 | 7 | 17 |
| 5 | **?** | 6 | 21 |

A) 10 B) 8 C) 5 D) 15

**Q9.** In each row, Top = Left + (2 × Middle) + Right. Work out the rule from the first two rows, then find the missing Middle number.

| Left | Middle | Right | Top |
|---|---|---|---|
| 2 | 3 | 4 | 12 |
| 5 | 2 | 6 | 15 |
| **?** | 5 | 3 | 20 |

A) 3 B) 20 C) 12 D) 7

**Q10.** In this grid, the product of the top-left and bottom-right numbers equals the product of the top-right and bottom-left numbers. Find the missing number.

| | |
|---|---|
| 4 | 6 |
| 8 | **?** |

A) 14 B) 2 C) 17 D) 12

**Q11.** This time look DOWN each column, not across each row. Every column adds up to the same total. Find the missing number.

| | | |
|---|---|---|
| 10 | 12 | **?** |
| 8 | 7 | 12 |
| 9 | 8 | 10 |

A) 5 B) 4 C) 12 D) 7

**Q12.** Each row follows a TWO-STEP rule — figure it out from the first two rows, then find the missing number in the third row.

| | | |
|---|---|---|
| 8 | 5 | 26 |
| 10 | 3 | 26 |
| 12 | 7 | **?** |

A) 40 B) 12 C) 36 D) 38

**Q13.** In this number pyramid, each number is the SUM of the two numbers directly below it. Find the missing number at the top.

```
          ?
     27     24
  12     15     9
```

A) 46 B) 54 C) 27 D) 51

**Q14.** Every row AND every column adds up to the same total. Find the missing number.

| | | |
|---|---|---|
| 9 | 8 | 5 |
| 6 | 10 | 6 |
| **?** | 4 | 11 |

A) 11 B) 9 C) 6 D) 7

**Q15.** Each row follows a TWO-STEP rule — figure it out from the first two rows, then find the missing number in the third row.

| | | |
|---|---|---|
| 4 | 5 | 7 |
| 6 | 8 | 10 |
| 5 | **?** | 9 |

A) 1 B) 6 C) 7 D) 11

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*Answer key:* `answer-key.md`

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